Task Parser Assistant

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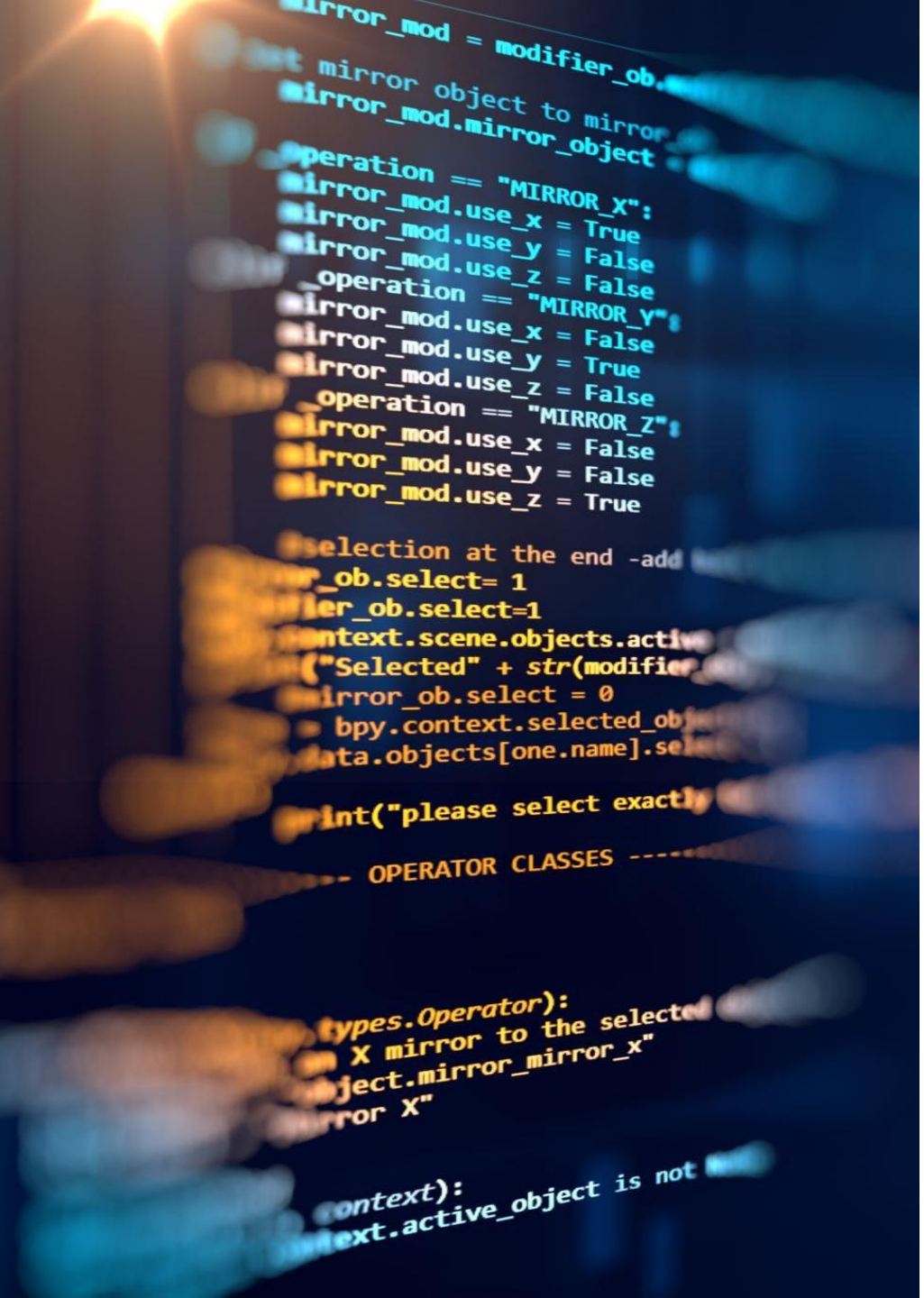
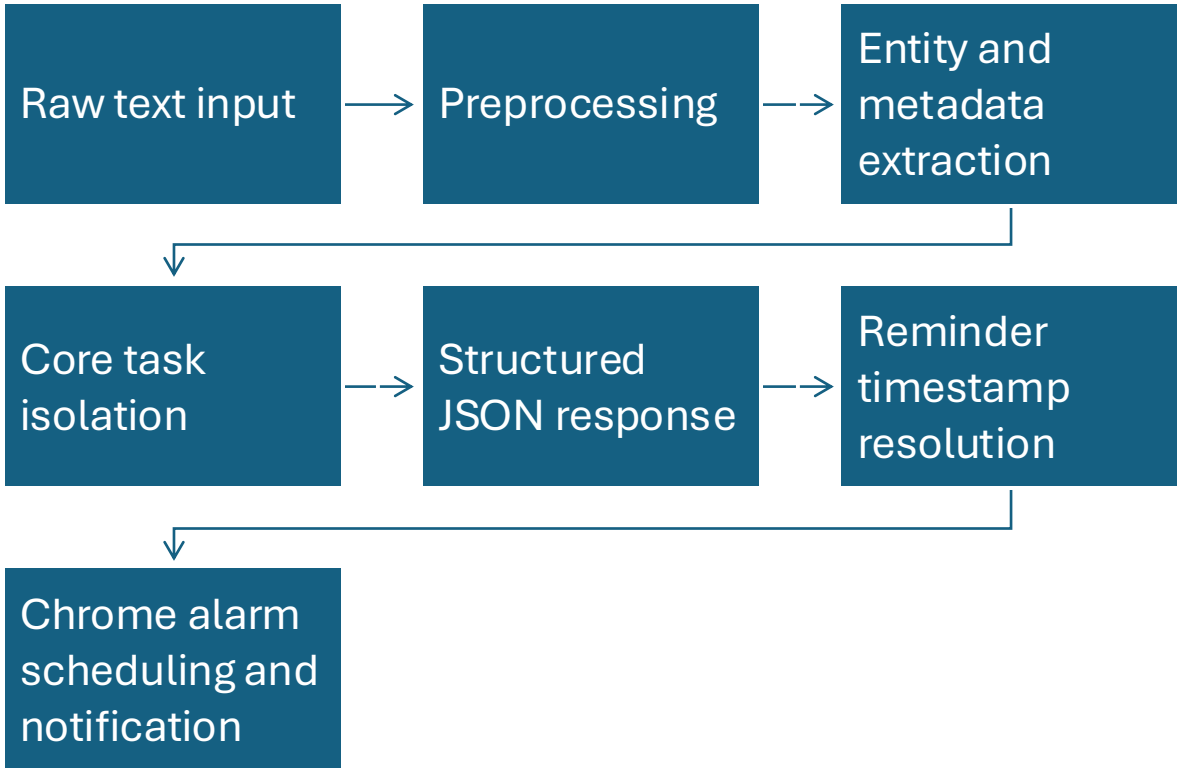
Project Overview

- Converts messy natural-language task input into structured JSON
- Supports preprocessing, entity extraction, task isolation, and confidence scoring
- Exposes parsing through a FastAPI backend
- Connects to a Chrome extension that can schedule browser reminders



Backend Pipeline



Preprocessing

Preprocessing – The Challenge

Human input is unpredictable: typos, slang, and abbreviations (e.g., "tmrw", "msg")

Standard spellcheckers destroy technical terms and formatting (e.g., "v3.2", "35lbs")

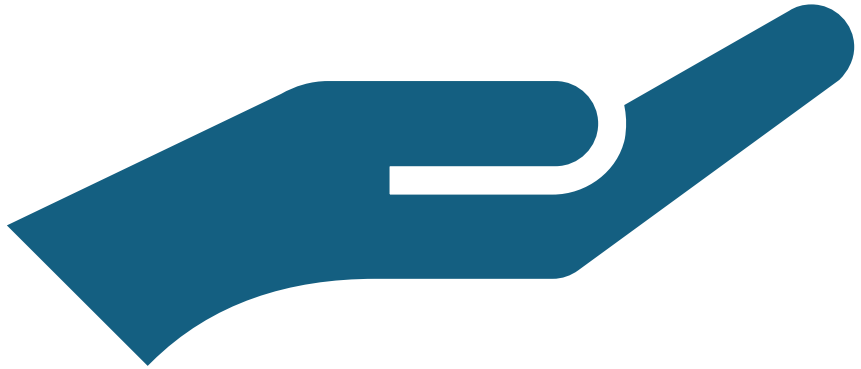
Lack of spaces between numbers and letter confuse tokenizers (e.g., "3pm")

Preprocessing – Our Solution

Built a domain-specific shorthand dictionary for productivity terms.

Implemented regex safeguards to protect acronyms (e.g., HDMI, FAQ, CEO) and decimals.

Applied selective spellchecking only to purely alphabetical tokens.



Entity Extraction

UTILIZED SPACY'S LLM FOR ACCURATE WORD VECTORS.

AUTOMATICALLY EXTRACTS STANDARD ENTITIES: DATE, TIME, PERSON, LOCATION.

ADDS DERIVED METADATA LIKE PRIORITY AND CATEGORY

REDUCES THE NEED FOR MANUAL REGEX PATTERN MATCHING FOR EVERY POSSIBLE NAME OR PLACE.

Task Extraction

- Goal: isolate the core action after metadata is identified
- Removes extracted date, time, and location fields from the cleaned sentence
- Also removes linked prepositions such as at, by, on, before, and in
- Cleans leftover punctuation and extra spaces
- Produces a shorter task string that is easier to store or pass into downstream systems
- Example:
  - Input: “Meet Sam at the library tomorrow at 3pm”
  - After entity removal: “Meet Sam”
  - Final task field: "Meet Sam"

Current Improvements / Updates

- FastAPI backend for structured parsing
- Chrome extension popup frontend
- Reminder timestamp resolution for vague time phrases
- Local Chrome alarm scheduling and notifications
- Confidence, warnings, and ambiguity tracking

Parser Output and Diagnostics

- Final output is returned as a structured JSON response
- Core Output:
  - task
  - date/time
  - person/location
  - command
  - priority /category
- Diagnostic fields:
  - confidence score
  - warnings
  - ambiguities
  - parser backend used
- Helps downstream systems decide whether a task is ready to schedule or needs user clarification


```
{
  "raw_text": "Remind me to meet Sam at 3pm tmr",
  "cleaned_text": "meet Sam at 3 pm tomorrow",
  "task": "meet Sam",
  "date": "tomorrow",
  "time": "3 pm",
  "person": [],
  "location": [],
  "command": "remind",
  "priority": "Normal",
  "category": null,
  "parser_backend": "spacy_blank_en",
  "confidence": 0.77,
  "warnings": [],
  "ambiguities": [
    "No strong category match was found."
  ]
}
```

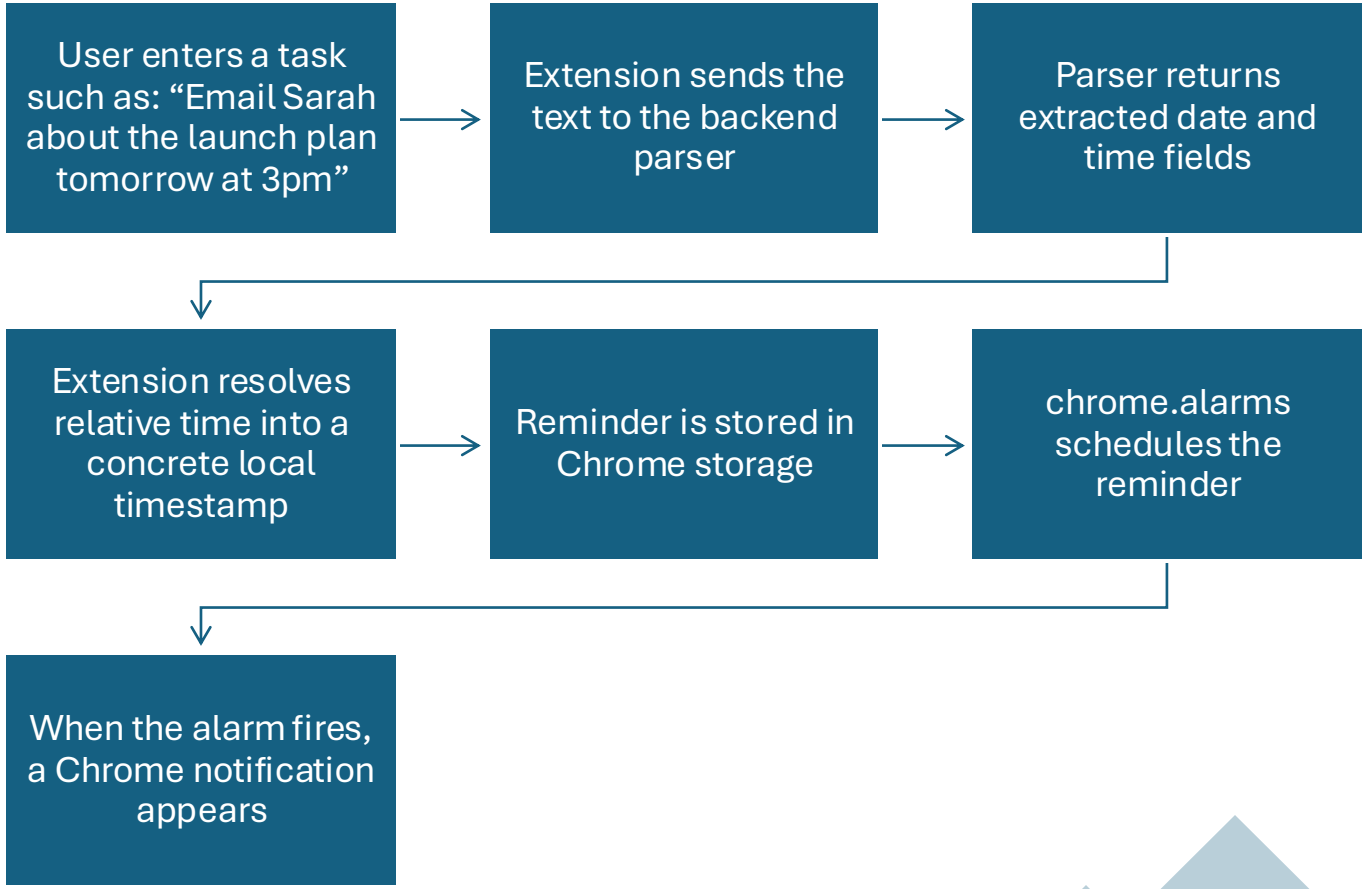
System Architecture

- User enters text in the Chrome popup
- Popup sends the text to the FastAPI `/parse` endpoint
- Backend runs preprocessing, extraction, and task isolation
- Backend returns structured JSON
- Popup resolves the final reminder timestamp
- Background script stores the reminder and schedules `chrome.alarms`
- Chrome notification appears when the alarm fires

Backend + Chrome Extension Integration

- FastAPI exposes a `/parse` API endpoint
- The popup sends raw task text to the backend
- The backend returns parsed fields in JSON
- The popup displays both a readable summary and raw output
- If the reminder time is valid, the extension can schedule it locally in Chrome

Reminder Workflow



Limitations and Future Work

- **Current limitations**
  - Some vague phrases still require interpretation rules outside the parser itself
  - Multiple people or locations can introduce ambiguity
  - Extraction quality depends on available NLP dependencies
  - Some environments may fall back to lighter parsing behavior
- **Future work**
  - Improve multi-entity disambiguation
  - Add editable backend URL / deployment support in the extension
  - Expand supported task categories and commands
  - Improve timestamp resolution for phrases like “later” or “soon”
  - Add cloud sync or cross-device reminder support

Demo